BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

October 25, 2026

Question 1

1a)

Yes.

[Definition of Complete Markets will be here]

Let

$$X = \begin{pmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{pmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$). Then there exists Ω such that $X\Omega = I$, with $\Omega = X^{-1}$, if X is invertible. Therefore, checking whether X is invertible is sufficient for Complete Markets in this economy. X has full rank. Therefore, this market is complete.

1b)

Let $P_c(n)$ denote the price of the contingent claim for state n. Then

$$P(A) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_A(n),$$
$$P(B) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_B(n),$$
$$P(C) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_C(n).$$

Or, in matrix form,

$$P = XP_c,$$

where X is the payoff matrix from 1a), $P_c = \big(P_c(1), P_c(2), P_c(3)\big)'$, and $P = \big(P(A), P(B), P(C)\big)' = (16, 8, 6)'$.

We confirmed in 1a) that the payoff matrix is invertible. So, we can calculate

$$P_c = X^{-1}P = \textit{[numerical values to be computed by the student]}.$$

1c)

As calculated in 1b), P_c = *[values from 1b)]*. Taking the inner product of $\text{Payoff}(D) = (30, 20, 10)'$ with P_c gives the price of asset D in this economy:

$$P(D) = \text{Payoff}(D)'\, P_c = \textit{[to be computed by the student]}.$$

1d)

The SDF M satisfies $\mathbb{E}[M \cdot \text{Payoff}] = \text{Price}$ for every asset in this economy:

$$\begin{aligned} P(A) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_A(n), \\ P(B) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_B(n), \\ P(C) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_C(n). \end{aligned}$$

Or, in matrix form,

$$P = X_\pi M, \qquad X_\pi = \begin{pmatrix} 10 \cdot \frac{1}{4} & 20 \cdot \frac{1}{2} & 30 \cdot \frac{1}{4} \\ 10 \cdot \frac{1}{4} & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \\ 0 & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \end{pmatrix},$$

where $M = \big(M(1), M(2), M(3)\big)'$ and $P = \big(P(A), P(B), P(C)\big)'$.

X_π is invertible if $\det(X_\pi) \neq 0$: $\det(X_\pi)$ = *[to be computed by the student]*. Therefore, the SDF realizations are

$$M = X_\pi^{-1} P = \textit{[numerical values to be computed by the student]}.$$

This economy satisfies No-Arbitrage.

[Definition of No-Arbitrage will be here]

As in 1b), every contingent claim satisfies the No-Arbitrage condition above.

As we can see, the realized SDF is the same in every state of the economy. It may imply that each state is equally valuable *[?]*.

1e)

Using the SDF from 1d), with $\text{Payoff}_D(1) = 30$, $\text{Payoff}_D(2) = 20$, and $\text{Payoff}_D(3) = 10$,

$$P(D) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_D(n) = \textit{[to be computed by the student; compare with 1c)]}.$$

Question 2

2a)

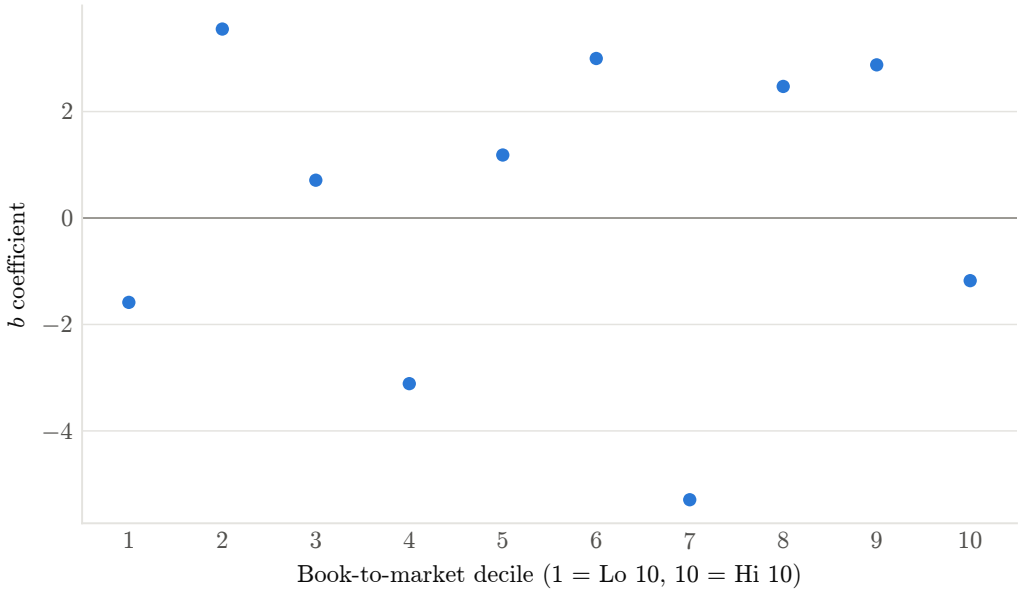


Figure 1: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted book-to-market decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2b)

The sample (Pearson) correlation between the monthly SDF series $M^*_{\mathrm{BM},t}$ estimated in 2a) and $M^*_{\mathrm{MOM},t}$ estimated in 2b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\mathrm{Corr}}\left(M^*_{\mathrm{BM},t},\, M^*_{\mathrm{MOM},t}\right) = 0.355.$$

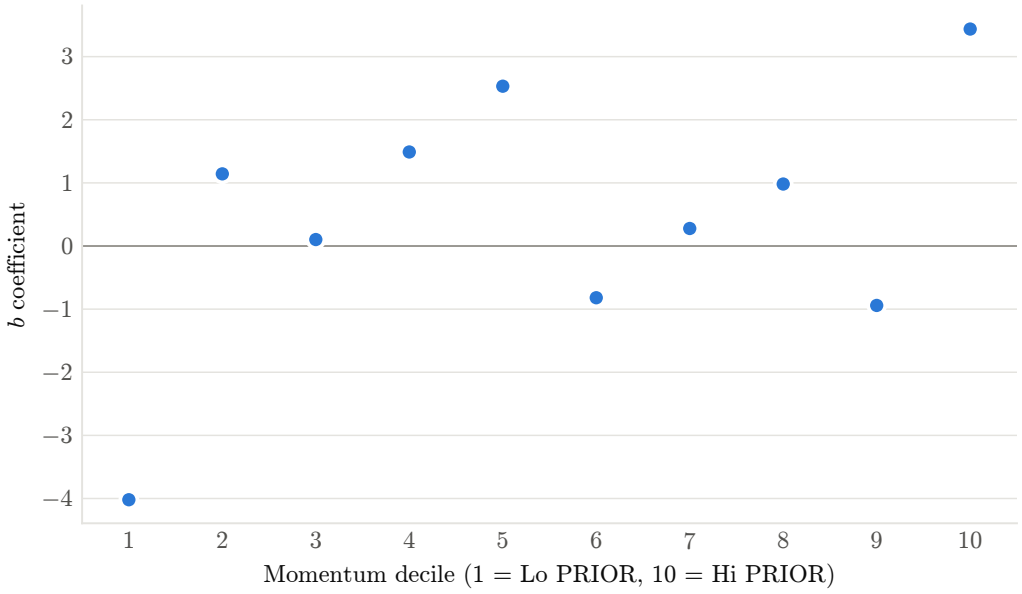


Figure 2: Coefficients b of the SDF $M_t^* = a - b' x R_t$ that prices the monthly excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2c)

Table 1: Monthly Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = b' x R_t$ implied by the SDFs of 2a)–2c), July 1927 to August 2026 ($T = 1190$ months). Not annualized.

SDF	Test assets	$\widehat{SR}(xR^*)$
2a)	10 book-to-market deciles	0.192
2b)	10 momentum deciles	0.258
2c)	20 book-to-market and momentum deciles	0.305

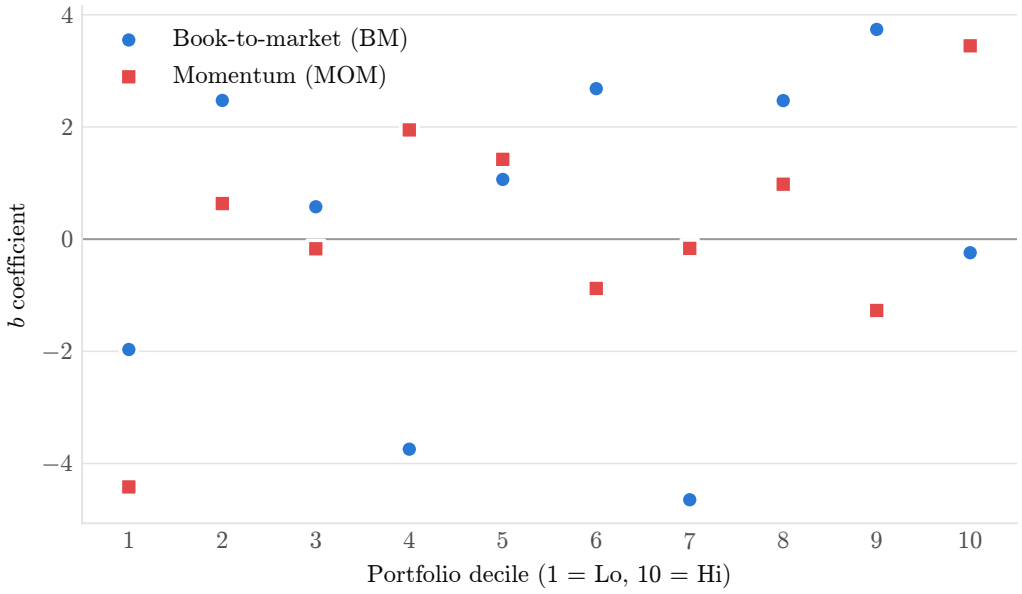


Figure 3: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 20 value-weighted portfolios in the merge of the book-to-market (blue circles) and momentum (red squares) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2d)

[Explanation will be here]

	SDF M_t^* estimated in		
Portfolio	2a) BM	2b) MOM	2c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.00	−0.15	0.00
2	0.00	−0.01	0.00
3	0.00	−0.02	0.00
4	0.00	−0.08	0.00
5	0.00	0.05	0.00
6	0.00	0.13	0.00
7	0.00	0.03	0.00
8	0.00	0.24	0.00
9	0.00	0.34	0.00
10 (Hi 10)	0.00	0.32	0.00
Panel B: momentum deciles			
1 (Lo PRIOR)	−1.02	0.00	0.00
2	−0.51	0.00	0.00
3	−0.33	0.00	0.00
4	−0.15	0.00	0.00
5	−0.09	0.00	0.00
6	−0.03	0.00	0.00
7	0.05	0.00	0.00
8	0.19	0.00	0.00
9	0.22	0.00	0.00
10 (Hi PRIOR)	0.54	0.00	0.00

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in 2a) (10 book-to-market deciles), 2b) (10 momentum deciles) and 2c) (all 20 portfolios), in percent per month. Monthly excess returns, July 1927 to August 2026 ($T = 1190$ months); covariances divide by T.

2e)

[Explanation will be here]

Portfolio	$\hat{\alpha}_{OS,p}$
Panel A: book-to-market deciles	
1 (Lo 10)	1.10
2	0.79
3	0.90
4	0.70
5	0.80
6	0.72
7	0.70
8	0.59
9	0.83
10 (Hi 10)	0.97
Panel B: momentum deciles	
1 (Lo PRIOR)	2.71
2	1.65
3	1.39
4	1.21
5	0.92
6	0.83
7	0.76
8	0.64
9	0.43
10 (Hi PRIOR)	0.55

Table 3: Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ of the 20 book-to-market and momentum decile portfolios, in percent per month. $\hat{a}$ and $\hat{b}$ are estimated as in 2c) on July 1927 to December 1989 ($T = 750$ months); $M^*_{OS,t} = \hat{a} - \hat{b}' xR_t$ and the moments are computed on January 1990 to August 2026 ($T = 440$ months); covariances divide by T.

Question 3

3a)

From the maximization problem in Equation (3.2),

$$\varpi_t^* = \arg\max_{\varpi_t} \mathbb{E}_t\big[U(W_{t+1})\big] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot \big(R_{f,t+1} + \varpi_t' xR_{t+1}\big),$$

where $R_{w,t+1} = R_{f,t+1} + \varpi_t^{*\prime} xR_{t+1}$ is the return on the optimal portfolio, so that $W_{t+1} = W_t \cdot R_{w,t+1}$.

The first-order condition is

$$\frac{\partial\, \mathbb{E}_t[U(W_{t+1})]}{\partial \varpi_t} = \mathbb{E}_t\big[U'(W_{t+1}) \cdot W_t \cdot xR_{t+1}\big] = \mathbf{0} \quad \Longleftrightarrow \quad \mathbb{E}_t\big[U'(W_{t+1}) \cdot xR_{t+1}\big] = \mathbf{0}.$$

Since W_t is the investor's wealth at time t, it is positive ($W_t > 0$) and known at time t, so it can be taken outside $\mathbb{E}_t[\cdot]$ and divided out.

Because $U'(W_{t+1}) = W_{t+1}^{-\gamma} = (W_t \cdot R_{w,t+1})^{-\gamma}$,

$$\mathbb{E}_t\big[(W_t \cdot R_{w,t+1})^{-\gamma} \cdot xR_{t+1}\big] = \mathbf{0} \iff W_t^{-\gamma}\,\mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot xR_{t+1}\big] = \mathbf{0} \iff \mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot xR_{t+1}\big] = \mathbf{0},$$

where the last step uses the same argument for $W_t^{-\gamma} > 0$.

Therefore, $M_{t+1} = R_{w,t+1}^{-\gamma}$ prices the vector of excess returns. Because an SDF that prices only excess returns still does so after multiplication by any positive factor (here, $W_t^{-\gamma}$), the scale of M is not identified; a normalization is therefore needed and is innocuous.

3b)

	GMM
$\hat{\gamma}$	1.887
	(0.593)
Newey–West lag	23
Sample	Jun 1928 – Dec 2021
T (months)	1123

Table 4: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, with $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset (annual returns observed monthly). Newey–West standard error in parentheses (Bartlett kernel, Newey–West (1994) automatic lag, rounded down).

3c)

	Estimate	s.e.
$\hat{\gamma}$	3.013	0.716
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	−0.0040	−0.86
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0019	0.86
$\chi^2(1)$	0.75	
p-value	0.388	
Newey–West lag	21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 5: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$ with an identity weighting matrix ($\hat{A} = \hat{G}'$). $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset (annual returns observed monthly); $z_{t-1} = \widehat{\mathbb{E}}^{OS}[xR_e]/\sigma_e^2$, where $\widehat{\mathbb{E}}^{OS}[xR_e]$ is the Problem Set 1 (2e) out-of-sample forecast made twelve months before t (the forecast of $xR_{w,t}$) and σ_e^2 is the variance of $R_{e,t}$ over the sample. Standard error and t-statistics use Newey–West (Bartlett kernel, Newey–West (1994) automatic lag, rounded down). χ^2: test that the two pricing errors are jointly zero, using the Moore–Penrose pseudo-inverse of their (rank-one) covariance matrix.

3d)

	Estimate	s.e.
$\hat{\gamma}$ (two-step)	2.956	0.706
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	-0.0022	-0.88
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0058	0.88
$\chi^2(1)$	0.77	
p-value	0.379	
Newey–West lag, $\hat{S}_1$ / $\hat{S}_2$	21 / 21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 6: Two-step GMM estimate of γ from the same two moment conditions as in 3c), $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$. Step 1 is the 3c) estimate ($\hat{W} = I$); step 2 uses $\hat{W} = \hat{S}_1^{-1}$, with $\hat{S}_1$ the Newey–West estimate at the step-1 estimate, and is not iterated further. Standard error, t-statistics and χ^2 use the general sandwich formulas with $\hat{A} = \hat{G}'\hat{S}_1^{-1}$, $\hat{G}$ at the step-2 estimate, and $\hat{S}_2$, the Newey–West estimate re-computed at the step-2 estimate (Bartlett kernel, Newey–West (1994) automatic lag, rounded down). χ^2: test that the two pricing errors are jointly zero, using the Moore–Penrose pseudo-inverse of their (rank-one) covariance matrix. Data, instrument and sample as in 3c).

Question 4

4a)

Conditionally linear CAPM. Let $M_{t+1} = M(F_{t+1}; \theta)$. It can be approximated by a Taylor expansion around $\mathbb{E}_t[F_{t+1}]$:

$$M(F_{t+1};\theta) = \sum_{n=0}^{\infty} \frac{M^{(n)}\big(\mathbb{E}_t[F_{t+1}];\theta\big)}{n!}\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big)^n.$$

Terms of second order and above can be neglected. Then,

$$M(F_{t+1};\theta) \approx M\big(\mathbb{E}_t[F_{t+1}];\theta\big) + M'\big(\mathbb{E}_t[F_{t+1}];\theta\big)\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big).$$

If we plug in $M(F_{t+1};\theta) = F_{t+1}^{-\gamma}$ with $F_{t+1} = R_{w,t+1}$, then

$$M_{t+1} \approx -\gamma\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1} + (\gamma+1)\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma}.$$

Therefore,

$$a_t = (\gamma+1)\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma}, \qquad b_t = \gamma\, \mathbb{E}_t[R_{w,t+1}]^{-\gamma-1}.$$

Disadvantage relative to $M_{t+1} = R_{w,t+1}^{-\gamma}$.

1. The original model, $M_{t+1} = R_{w,t+1}^{-\gamma}$, contains only the factor and the parameter, so there is no concern about additional assumptions or time-varying coefficients.

2. After linearization, a_t and b_t become time-varying, so $\mathbb{E}[xR_j]$ can no longer be written in the form $\mathbb{E}[b_t]' \operatorname{Cov}(F_t, xR_{j,t})$.

3. The time variation in a_t and b_t comes from $\mathbb{E}_t[R_{w,t+1}]$, as shown above. This conditional expectation must be estimated, and the choice of how to estimate it (e.g., with a linear model or some other method) introduces estimation error. The model could be wrong, or there could be estimation error even if the model is correct.

4. By imposing additional assumptions or a model on a_t and b_t, they can be made time-invariant. However, if that model is wrong or the assumptions are inconsistent with reality, the estimates move further away from reality.

5. $\mathbb{E}_t[xR_{t+1}] = b_t' \operatorname{Cov}_t(F_{t+1}, xR_{t+1})$ holds, but from the perspective of the unconditional implications used in GMM, it can be written as $\mathbb{E}[xR_j] = \mathbb{E}\big[b_t' \operatorname{Cov}_t(F_{t+1}, xR_{j,t+1})\big]$.

6. But again, b_t is determined by $\mathbb{E}_t[R_{w,t}]$. As in Problem Set 1, we already checked that estimating the equity premium *very well*, either in sample (IS) or out of sample (OS), is really difficult.

4b)

	Estimate	s.e.
$\hat{b}$	1.975	0.605
	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Book-to-market deciles		
1 (Lo 10)	−0.0062	−0.56
2	0.0163	1.98
3	0.0054	0.80
4	−0.0017	−0.28
5	0.0109	1.60
6	0.0163	2.12
7	−0.0014	−0.15
8	0.0195	2.34
9	0.0266	2.56
10 (Hi 10)	0.0083	0.56
Momentum deciles		
1 (Lo PRIOR)	−0.1110	−8.76
2	−0.0493	−5.45
3	−0.0275	−3.00
4	−0.0128	−1.84
5	0.0000	0.00
6	0.0003	0.06
7	0.0163	2.16
8	0.0298	4.57
9	0.0360	4.02
10 (Hi PRIOR)	0.0709	5.10
Market		
xR_w	−0.0002	−0.03
$\chi^2(20)$	229.37	
p-value	<0.001	
Newey–West lag	22	
Sample	Jun 1928 – Dec 2021 ($T = 1123$)	

Table 7: First-stage GMM estimate of b in $M_t = 1 - b\,(R_{w,t} - \mu)$, $\mu = \mathbb{E}[R_w]$, from $\widehat{\mathbb{E}}[M_t \cdot xR_t] = 0$ with identity weights on the 21 pricing moments ($\hat{b} = (\hat{d}'\hat{d})^{-1}\hat{d}'\,\widehat{\mathbb{E}}[xR]$, $\hat{d} = \widehat{\mathrm{Cov}}[R_w, xR]$) and the moment $\mathbb{E}[R_w - \mu] = 0$ satisfied exactly. xR stacks trailing 12-month excess returns of the 20 book-to-market and momentum decile portfolios and $xR_w = e^{r_e} - e^{r_f}$ (EQ Dataset), in decimals, observed monthly; $R_w = e^{r_e}$. Standard errors and t-statistics use Newey–West (Bartlett kernel, Newey–West (1994) automatic lag) and account for the estimation of μ. χ^2: test that the 21 pricing errors are jointly zero, using the Moore–Penrose pseudo-inverse of their covariance matrix.